# Clustering Workers and Firms:
## Types and Match Qualities

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November 27, 2025

# Research Questions

- How to **identify types of workers/firms** from the employment/wage history,
  **without assuming ranking**?
  - **Clustering (e.g., Netflix's Machine learning algorithm)**
- How to cluster workers?
  - **Use the similarity between each pair of workers**:
    have **worked in the same (types of) firms** (at similar frequencies)
    have **received similar wages** in the same types of firms
  - Cluster workers with high similarities into the same cluster
  - So-called spectral clustering

# Plan of the Talk

- Setup of the problem
- Algorithm: Compared to Netflix Algorithm
- Results of implementing the algorithm into Simulated Data

# Setup of the Problem I

- Discrete time $t \in \{1, 2, \dots T\}$
- A worker $i$ belongs to a cluster $c$
  - In total $I$ workers and $N_c$ worker clusters
- A firm $j$ belongs to a cluster (or group) of firms $g$
  - In total $J$ firms and $N_g$ firm clusters

# Setup of the Problem II

- Workers from the same cluster are ex-ante identical, and so are firms. Implications:
  - **Productivity** of worker $i$ in firm $j$ depends only by the types of the worker ($c_i$) and of the firm ($g_j$)

$$y(i,j,t) = f_y(c_i, g_j).$$

  - **Real wage** also depends only on the types

$$w^r(i,j,t) = f_w(c_i, g_j),$$

    determined by output and bargaining

- **Observed wage** equals real wage plus noise

$$w(i,j,t) = w^r(i,j,t) + \varepsilon(i,j,t).$$

# Question

- We observe
  - Employment history of each worker $i$: $i, t, j(i,t) \in \{0, 1, 2, ..., J\}$[1]
  - **Observed wage**: $w(i,j,t)$.
  - **Observed tenure**: how long you stay in a firm.
  - From simulated data using a search-matching model, e.g., HLM (2017),
  - or the real world data (Norwegian registration data)?
- We want to
  - Identify types of workers $c_i$ and firms $g_j$

[1] $j(i,t) = 0$ means that $i$ is unemployed.

# The Netflix Analogy: A Familiar Problem

### Netflix Recommendation

**Problem:** Cluster users and movies, back out unobservable types of users: which movies do you really like

- Users watch movies
- Observe: watch time + ratings
- Similar users → same cluster

### Labor Market

**Problem:** Cluster workers and firms, back out unobservable types of workers and firms

- Workers employed by firms
- Observe: tenure + wages
- Similar workers → same cluster

**Same principle: collaborative filtering**

# Netflix Algorithm: The Old Approach

## Traditional Recommendation (Pre-2006)

**Use observables, linear regressions to predict popularity of movies**

- Users rate movies: 1-5 stars
- User and movie characteristics: observables
- Algorithm predicts ratings for unwatched movies
- Recommend highest predicted ratings

## The Problem: Stated vs. Revealed Preferences

**Example:** A university professor rates romantic comedies 2 stars ("I don't like rom-coms")

**But...** watch them for 3 hours!

*Stated* preference $\neq$ *True* preference
*Observed* characteristics may not be good enough predictors

# Netflix Prize Innovation: Revealed Preferences

## The Breakthrough (2006-2009)

**Use implicit behavior to back out true types and match quality**

- **Watch time:** How long did you actually watch?
- **Completion rate:** Did you finish the movie/series?

## Revealed Preference

3 hours watch
$\Rightarrow$ True match quality: HIGH
$\Rightarrow$ You're actually a rom-com fan!

## Stated Preference

High rating + 10 minutes watch
$\Rightarrow$ True match quality: LOW
$\Rightarrow$ Just trying to look intellectual

**Result:** Netflix originals based on revealed preferences:
*House of Cards*, *Love is Blind*, *Squid Game*

# Mathematical Framework

## Netflix Similarity

For users $u, u'$ who watched movie $m$:

$$s_{\text{Netflix}}(u, u', m) = f\,(rating, time)$$

## Labor Market Similarity

For workers $i, i'$ employed by firm $j$:

$$s_{\text{Labor}}(i, i', j) = f\,(wage, tenure)$$

**Isomorphic problem structure**

# Benchmark Algorithm to Identify Workers

Spectral clustering - see Luxburg (2007)

1. Computer worker $i$'s average wage in firm $j$

$$\bar{w}(i,j) = mean(w(i,j,t)), \forall t \text{ s.t. } j(i,t) = j.$$

2. **Compute similarity of workers** $i$ and $i'$ in firm $j$, if they have both worked in firm $j$

$$s(i,i',j) = \exp\left(-\frac{(\bar{w}(i,j) - \bar{w}(i',j))^2}{2\sigma^2}\right).$$

3. Compute similarity of worker $i$ and $i'$:

$$s(i,i') = \max_j s(i,i',j),$$

which gives the similarity matrix $S$:

$$S_{i,i'} = s(i,i').$$

4. **Implement k-means clustering to the similarity matrix** $S$ to cluster workers into $N_c$ clusters.

# Understanding the Similarity Matrix

- Transform the data from a $I \times J$ matrix to a $I \times I$ matrix

| $i \backslash j$ | 1 | 2 | ... | $J$ |
|---|---|---|---|---|
| 1 | $\bar{w}(1,1)$ | $\bar{w}(1,2)$ | ... | $\bar{w}(1,J)$ |
| 2 | $\bar{w}(2,1)$ | $\bar{w}(2,2)$ | ... | $\bar{w}(2,J)$ |
| ... | ... | ... | ... | ... |
| $I$ | $\bar{w}(I,1)$ | $\bar{w}(I,2)$ | ... | $\bar{w}(I,J)$ |

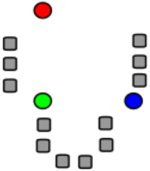
$\Rightarrow$

| $i \backslash i'$ | 1 | 2 | ... | $I$ |
|---|---|---|---|---|
| 1 | 1 | $s(1,2)$ | ... | $s(1,I)$ |
| 2 | $s(2,1)$ | 1 | ... | $s(2,I)$ |
| ... | ... | ... | ... | ... |
| $I$ | $s(I,1)$ | $s(I,2)$ | ... | 1 |

- Each worker is now a point in the $I$-dimension Euclidean space
    - Characterized by her similarities to other workers

# K-Means Clustering

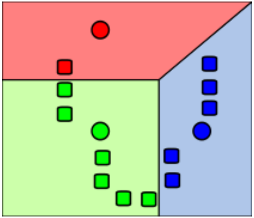
- Cluster $N$ points in the $M$ dimension Euclidean space into $C$ clusters
- Initialization: **pick a centroid for each cluster** (may not be a point of the data)



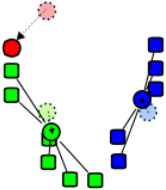
# K-Means Clustering

**Updating** until convergence of the centroids (two steps):

1. **Assign each point to the cluster of the nearest centroid**

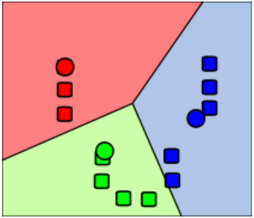


2. **Update the centroids** using the clusters of points



# K-Means Clustering

Keep updating...



# Implementation of the Benchmark Algorithm

- To simulated data using the model of HLM (2017)
- $T = 240$
- $I = 1000, N_c = 10$
- $J = 100, N_g = 10$, and each firm offers 10 jobs
- Positive assortative matching

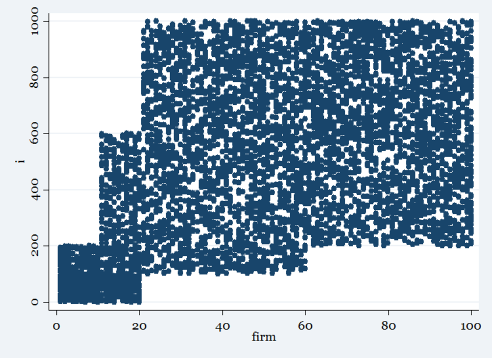
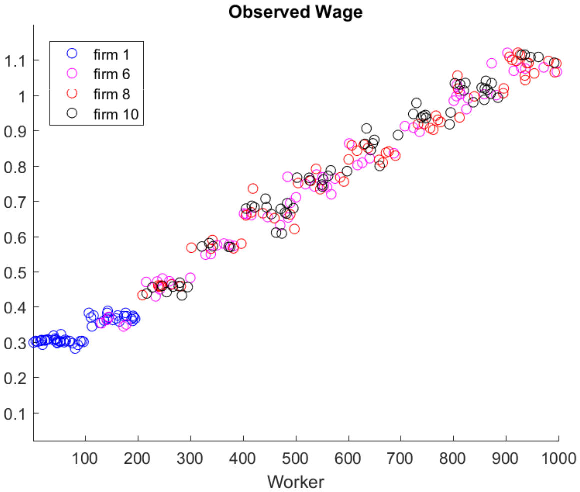



Figure: Workers Employed by Firms

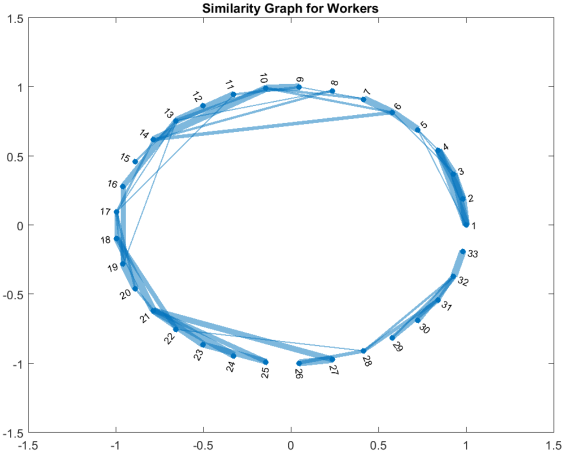
# Observed Wages



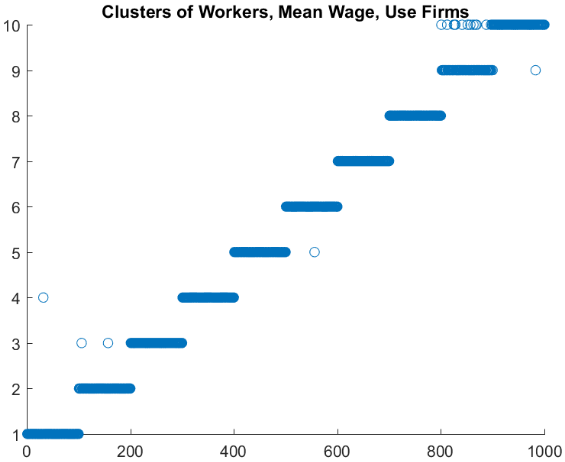

- Wage differences of different types of workers are clear
- Wage differences of different types of firms are not clear

# Similarity Graph of Workers

Pick 33 workers from 1000 workers, equally spaced



# Results: Worker Clusters



# Modifications of the Algorithm

- Use **firm cluster information** (will be available after clustering firms)
- Use **employment history information**: frequency/length of matching with a firm/firm cluster

$$s_f\left(i,i',j\right)=\exp\left(-\frac{\left(f\left(i,j\right)-f\left(i',j\right)\right)^2}{2\sigma_f^2}\right),$$

  and the final similarity can be a weighted sum of $s_f$ and $s$
- Use real wages
- Without constructing the similarity matrix (directly cluster workers using the wage matrix)
  - Does not work well
  - Distance of two workers is determined mainly by whether they work in the same firm:

$$w\left(i,j,t\right)-0\gg w\left(i,j,t\right)-w\left(i',j,t\right).$$

# Enhanced Algorithm with Tenure

Combine wage and employment duration:

### Wage Similarity

$$s^w\left(i,i',j\right) = \exp\left(-\frac{\left(\bar{w}\left(i,j\right) - \bar{w}\left(i',j\right)\right)^2}{2\sigma_w^2}\right)$$

### Tenure Similarity

$$s^f\left(i,i',j\right) = \exp\left(-\frac{\left(f\left(i,j\right) - f\left(i',j\right)\right)^2}{2\sigma_f^2}\right)$$

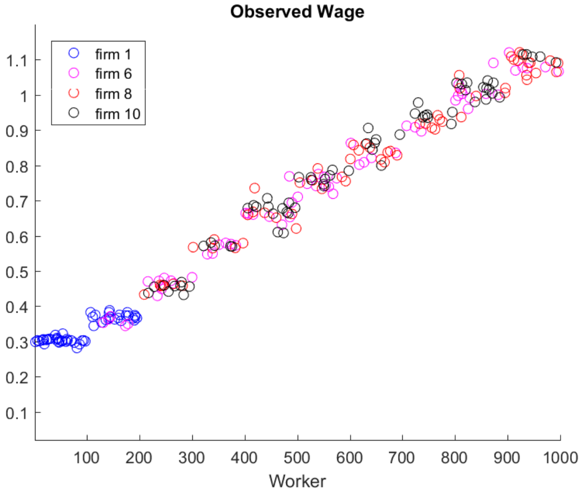
### Combined

$$s\left(i,i',j\right) = \alpha \cdot s^w + (1-\alpha) \cdot s^f$$

# Modifications of the Algorithm

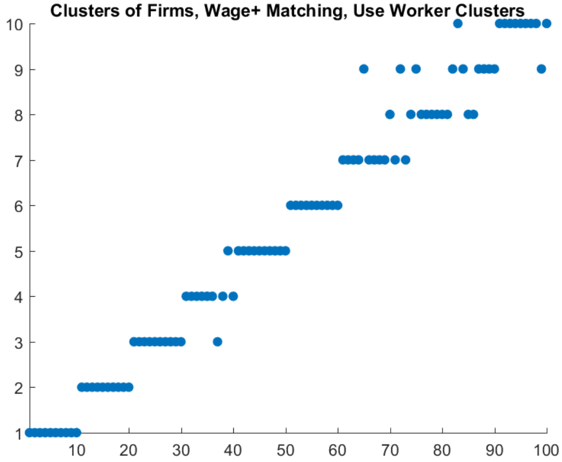
# Clustering Firms

- The same algorithms
- Difficulty: wages offered by two different types of firms to the same type of workers are not so different



# Clustering Firms

Necessary to use information on employment history and worker clusters



# Clustering Firms: Alternatives

# Iteratively Cluster Workers and Firms

1. Cluster workers, without knowing firm clusters
2. Cluster firms, knowing worker clusters
3. Cluster workers, knowing firm clusters
4. Iterate ...

Similar results as using the accurate clusters:
because **the first step provides quite accurate worker cluster results**

# Future Work

- Implement the algorithm to larger dataset (simulated/real data)
- Improve memory efficiency
  - **Reduce the dimension** of the similarity matrix
    $I$ dimension vectors can be represented by $N_c$ eigenvectors
  - Store a sparse similarity matrix
  - Compute similarity only when necessary
  - Automatically done by Python toolbox "scikit-learn"

# Application to Real Data

## Norwegian Registry Data

- Complete employment histories
- All Norwegian residents
- Detailed wage information
- Long time series

# Policy Applications

## Understanding Labor Markets

- Wage inequality decomposition
- Job mobility patterns
- Match quality identification

## Policy Evaluation

- Training program targeting

# Extension: Dynamic Types

**Current:** Static types throughout career

**Extension:** Allow types to evolve

- Workers gain experience
- Apply clustering within time windows
- Track type transitions over careers

**Implementation:**

- Divide career into phases (5-year windows)
- Cluster within each phase
- Construct transition matrices

# Summary

### Contributions

Novel non-parametric approach to identify types

### Key Insight

Netflix recommendation principles solve identification problems in labor market research

# Thank You!

Questions?